External side effects and recovery

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for External side effects and recovery before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Timeouts retries and cancellation

Without opening the old lesson, explain one key idea from Timeouts retries and cancellation. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Planning and action selection

Without opening the old lesson, explain one key idea from Planning and action selection. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Failure budgets and service objectives

Without opening the old lesson, explain one key idea from Failure budgets and service objectives. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Lesson reference for workbook practice

> Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

External effects can succeed while their **acknowledgement** is lost. Recovery must inspect committed state before repeating the operation.

The mock ledger records prepared, committed or unknown per operation ID. Retry a committed operation by returning its result; reconcile unknown status instead of assuming failure.

Preserve **compensation** as a separate authorised action.

Practise with fading help

> Complete the supported External side effects and recovery trace and record the first step you can explain without help.

Use W2: Trace the worked case for External side effects and recovery. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

WORKBOOK · 3 OF 15

Individual reasoning check

> Check the proposed External side effects and recovery explanation against the supplied facts before discussing it.

Use the worked case for External side effects and recovery. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply a new case independently

> A payment-like classroom token transfer has unknown status after network loss. Should the mock repeat immediately?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Explain the mechanism

> Explain the mechanism. Apply the supplied External side effects and recovery facts and keep your reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

W2 Trace the worked case

> Trace the worked case. Apply the supplied External side effects and recovery facts and keep your reasoning.

A booking is committed in storage, but response delivery times out. Caller asks to repeat. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

W3 Audit the limit

> Audit the limit. Apply the supplied External side effects and recovery facts and keep your reasoning.

A report omits this qualification: A compensating action is itself a side effect requiring authority; recovery does not silently permit deletion or refund.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied External side effects and recovery facts and keep your reasoning.

> A payment-like classroom token transfer has unknown status after network loss. Should the mock repeat immediately?

My reasoning and evidence

WORKBOOK · 9 OF 15

W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied External side effects and recovery facts and keep your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect External side effects and recovery to Grounded Learning Service, identify a limit and choose the next useful test.

Add a labelled external side effects and recovery **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of external side effects and recovery.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**
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WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Pictorial case practice: External side effects and recovery

> Recover an uncertain print job. A badge printer loses its reply after receiving a print request. Explain the labelled step before checking the changed condition.

Recover an uncertain print job. A badge printer loses its reply after receiving a print request.

Job key J6 identifies one badge. The printer supports status lookup by key; missing **acknowledgement** leaves outcome unknown.

AI ACADEMY / AI-6/17

External side effects and recovery

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Record uncertainty

Sending occurred, but printing completion is not yet known.



2. Query J6

Ask for the existing job's recorded state.



3. Recover from status

If complete, return that result without issuing another print job.

CHANGE ONE THING

Status reliably reports J6 absent.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Distinguish a missing printer **acknowledgement** from confirmed failure. What should happen if J6 is complete, and what **evidence** changes when J6 is reliably reported absent?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

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Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Grounded Learning Service record.

Project: Grounded Learning Service

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Recover an uncertain print job. A badge printer loses its reply after receiving a print request.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? If complete, return that result without issuing another print job.

1. Record uncertainty

2. Recover from status

3. Query J6

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Recover from status If complete, return that result without issuing another print job. A lost response does not show whether an external effect occurred.

Recovery distinguishes unknown outcome from a confirmed failed action.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Status reliably reports J6 absent. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. A policy-approved resend using the same deduplication key may be considered.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: A policy-approved resend using the same deduplication key may be considered. Confirmed absence supplies different **evidence** from silence.

Repair a missing building block

Review Timeouts retries and cancellation (ai-6/16). Goal: Explain the **mechanism** and terminology of timeouts retries and cancellation.

Short practice: Calculate waiting time for a one-second timeout followed by a 0.4-second success. What changes if cancellation arrives after the first attempt but before the retry?.

Return to this week and explain what you changed.

Review Planning and action selection (ai-6/14). Goal: Explain the **mechanism** and terminology of planning and action selection. Short practice: Choose a permitted plan when no approved tower-height card exists.

What new answer becomes supported after a card records 24 metres, and why could the bot not invent it earlier?. Return to this week and explain what you changed.

Review Failure budgets and service objectives (ai-6/11). Goal: Explain the **mechanism** and terminology of failure budgets and service objectives. Short practice: Calculate the maximum allowed failures among 100 requests.

Does 97 successes and 3 failures meet the target, and what changes after one failure becomes a success?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
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WORKBOOK · 15 OF 15

Visual concept journey

> Explain each External side effects and recovery picture from its labels, then record a relevant limit.

01. External side effects and recovery: the case

AI ACADEMY / AI-6/17 · 01 · LOOK AT THE FACTS

External side effects and recovery: the case

The situation

Recover an uncertain print job. A badge printer loses its reply after receiving a print request.

Supplied evidence

Job key J6 identifies one badge. The printer supports status lookup by key; missing acknowledgement leaves outcome unknown.

Explain external side effects and recovery using the deciding step in this fictional service.

Follow the labelled picture with the matching explanation. Open the vector diagram

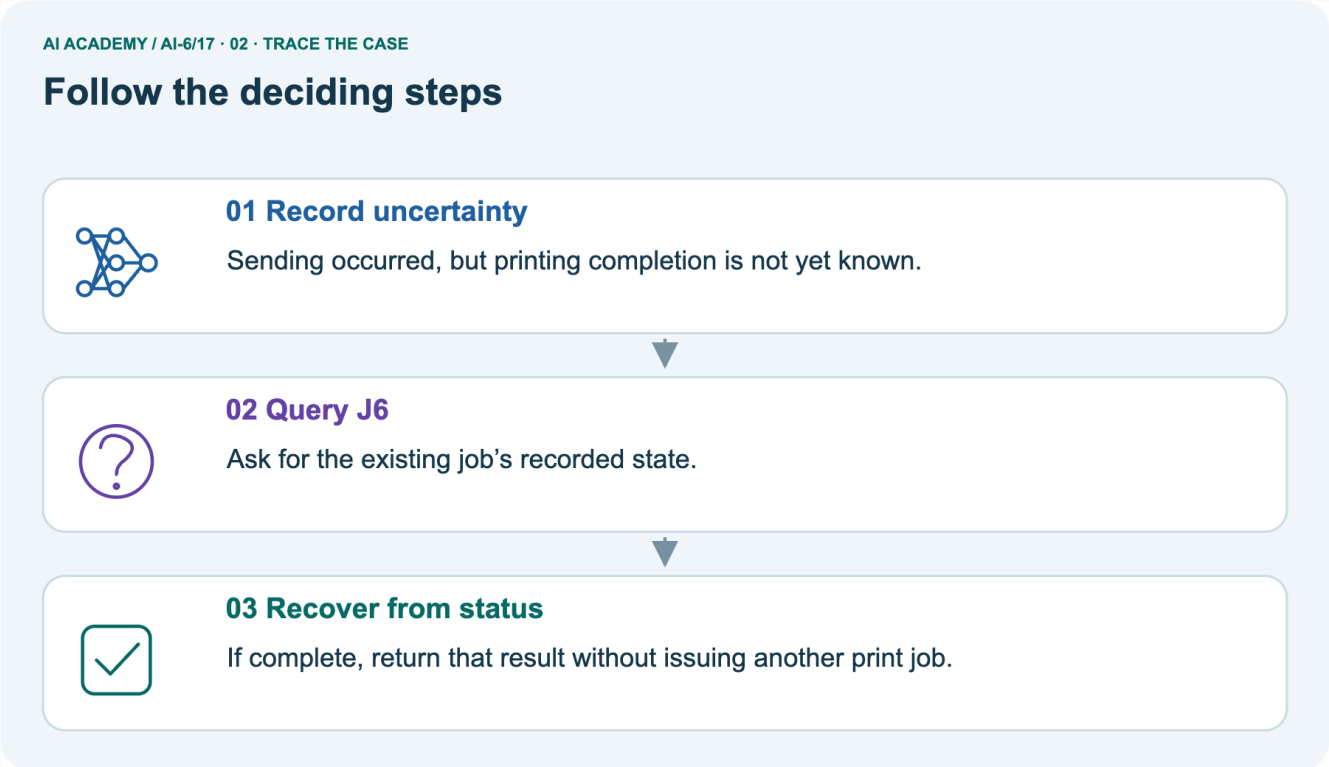
Recover an uncertain print job. A badge printer loses its reply after receiving a print request.
Supplied facts: Job key J6 identifies one badge.

The printer supports status lookup by key; missing **acknowledgement** leaves outcome unknown.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



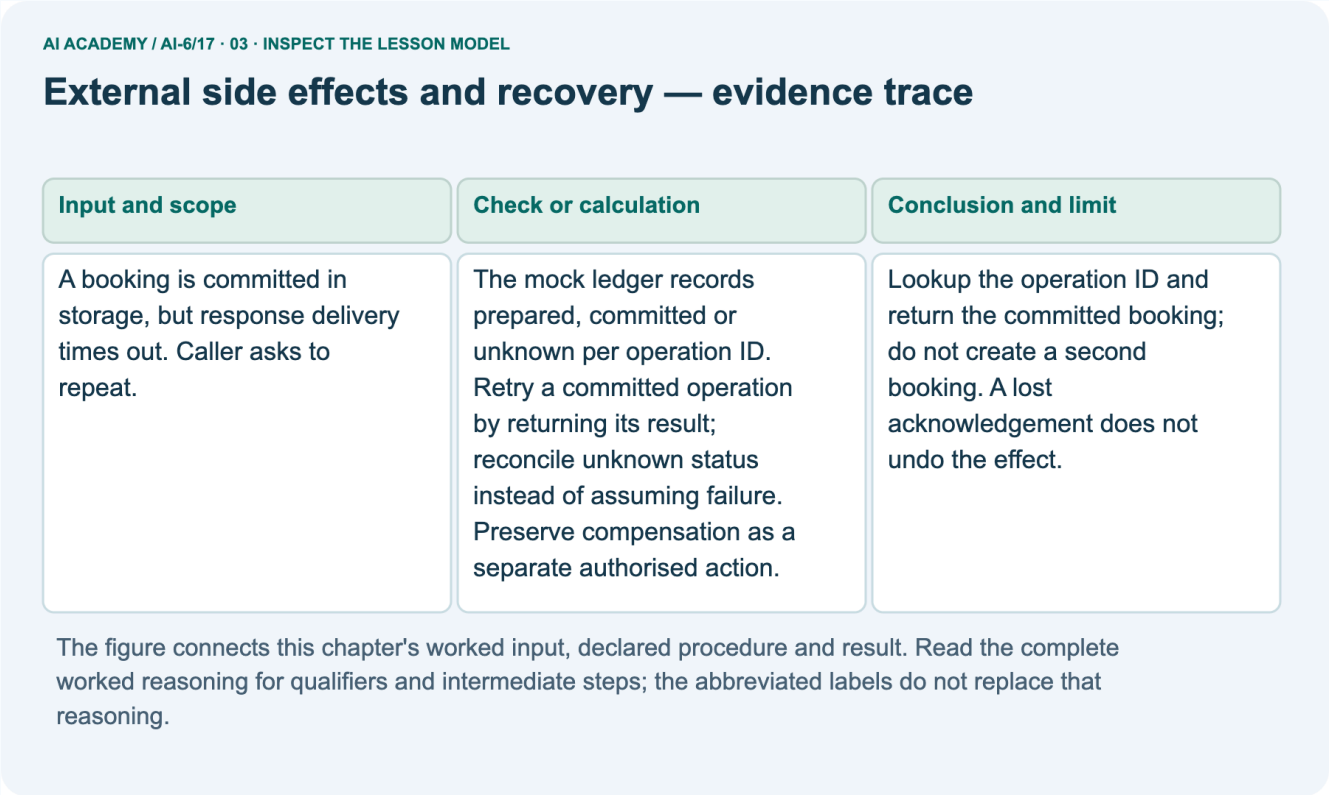
Follow the labelled picture with the matching explanation. Open the vector diagram

Record uncertainty: Sending occurred, but printing completion is not yet known. → Query J6: Ask for the existing job's recorded state. → Recover from status: If complete, return that result without issuing another print job.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. External side effects and recovery — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

External side effects and recovery — **evidence** trace. A booking is committed in storage, but response delivery times out. Caller asks to repeat. | The mock ledger records prepared, committed or unknown per operation ID.

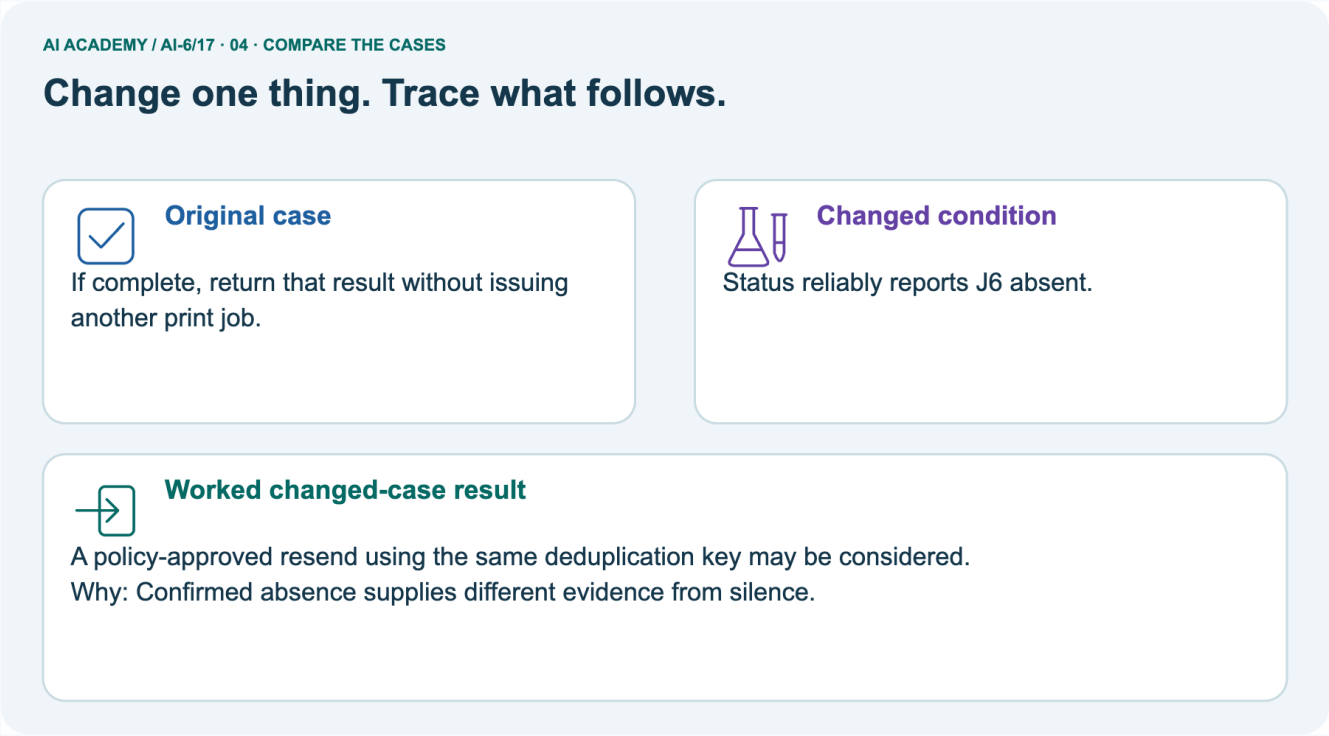
Retry a committed operation by returning its result; reconcile unknown status instead of assuming failure.

Preserve **compensation** as a separate authorised action. | Lookup the operation ID and return the committed booking; do not create a second booking. A lost **acknowledgement** does not undo the effect.

> Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

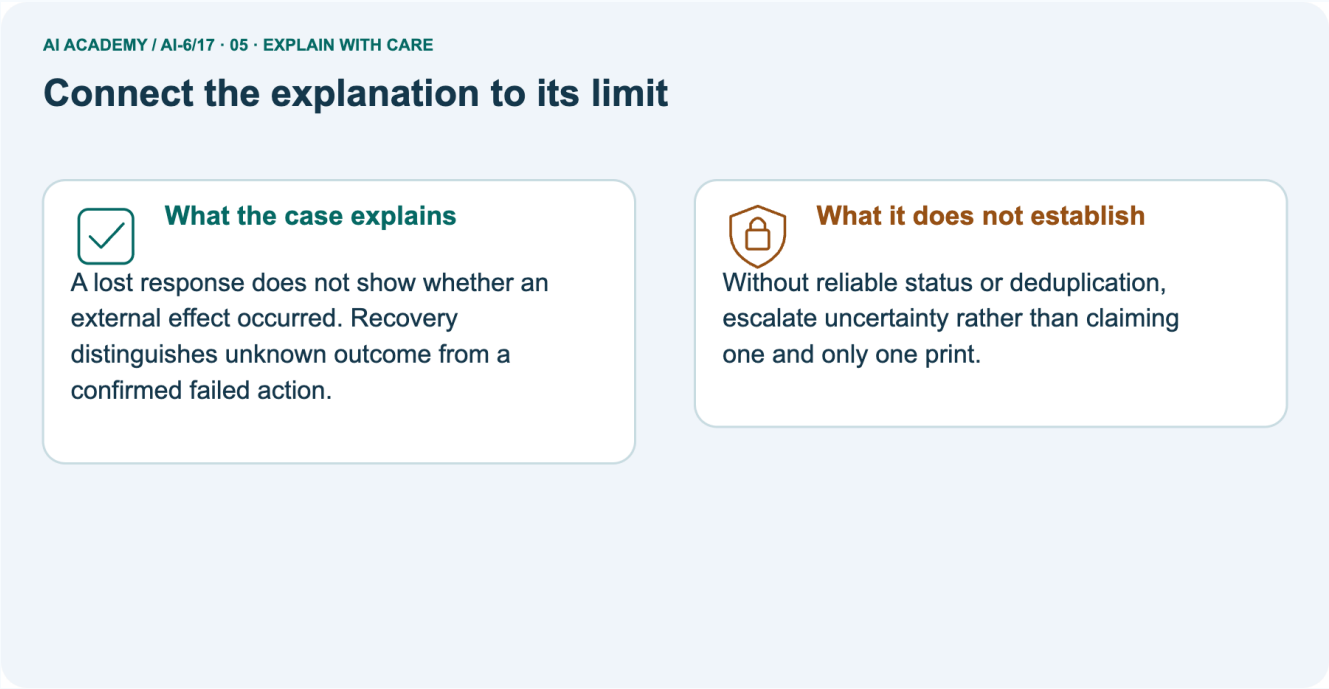
Original case: If complete, return that result without issuing another print job. Changed condition: Status reliably reports J6 absent.

Expected result: A policy-approved resend using the same deduplication key may be considered. Reason: Confirmed absence supplies different **evidence** from silence.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

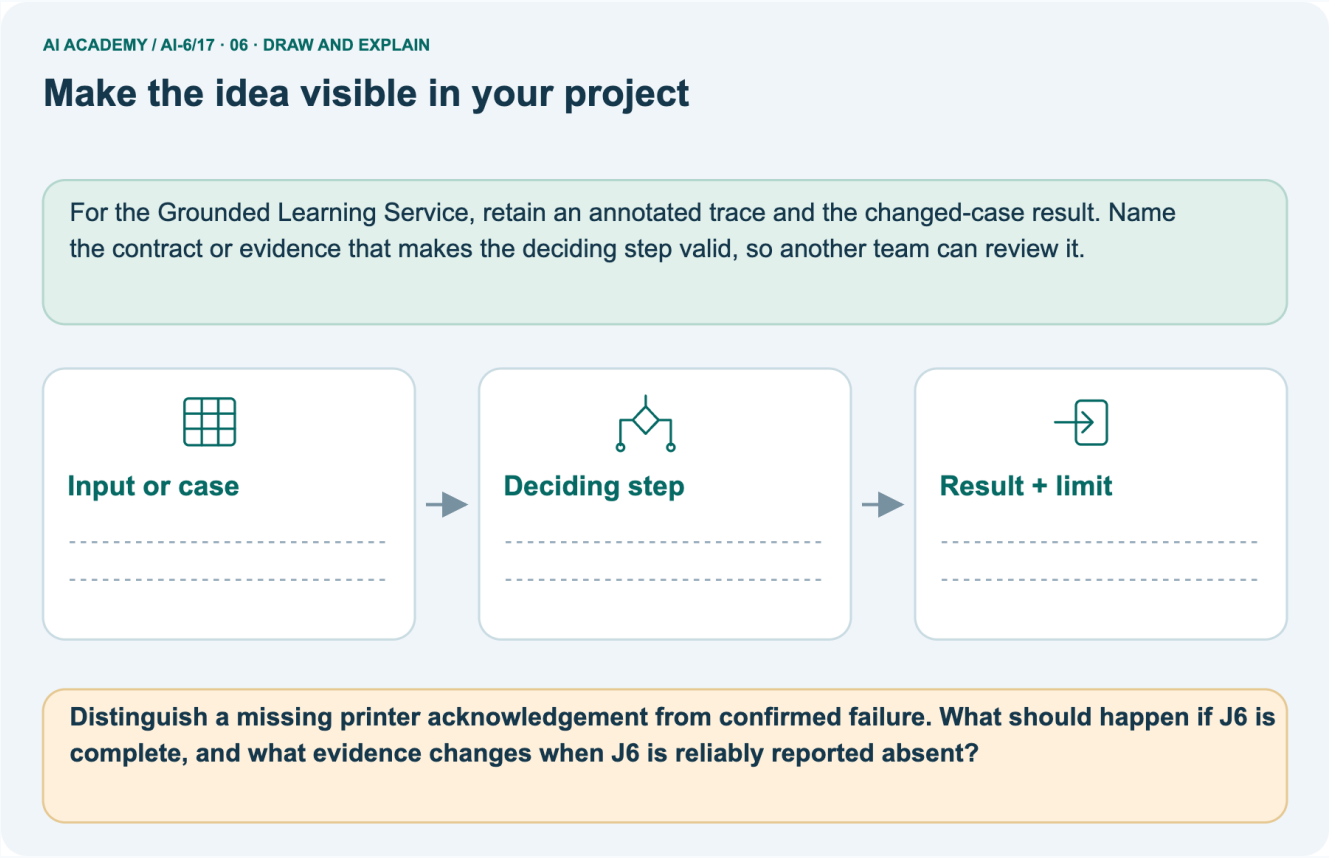
A lost response does not show whether an external effect occurred. Recovery distinguishes unknown outcome from a confirmed failed action.

Limit: Without reliable status or deduplication, escalate uncertainty rather than claiming one and only one print.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Distinguish a missing printer **acknowledgement** from confirmed failure.

What should happen if J6 is complete, and what **evidence** changes when J6 is reliably reported absent?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.